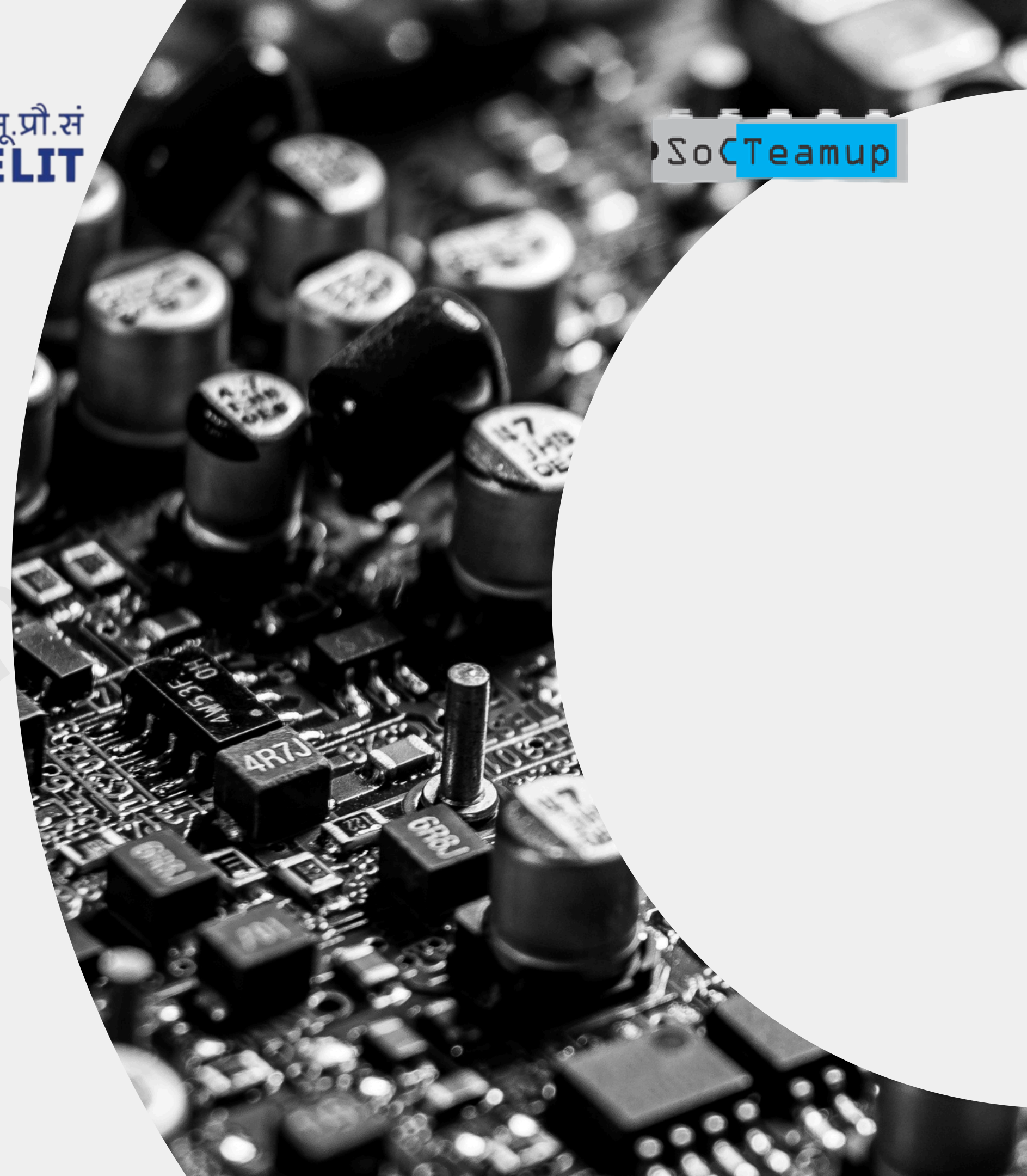


SOC Level Consideration





Outline

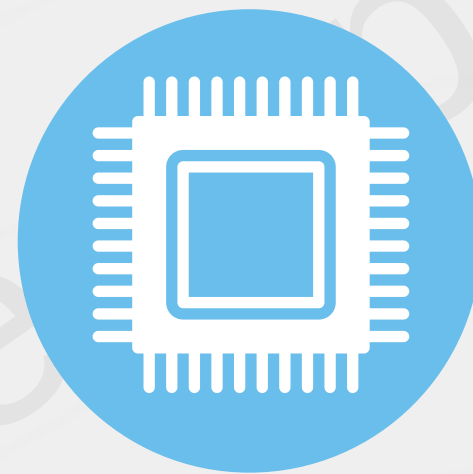
What Makes SoC Design Unique?
SoC Clock Domain Planning
Why Multiple Clock Domains Are Used
Clock Domain Crossing Challenges Recap
Reset Domain Partitioning and Design
Global vs Local Reset Control
Low-Power Design: Challenges and Goals
Dynamic Power vs Static (Leakage) Power
Low-Power Design Methodologies (UPF/CPF)
Summary of SoC-Level Best Practices

➔ What Makes SoC Design Unique?

- SoC-> Multiple IPs integrated: CPU, Memory, DSP, IOs, etc.
- Each IP may have its own clock, reset, power domain
- So, Integration needs:

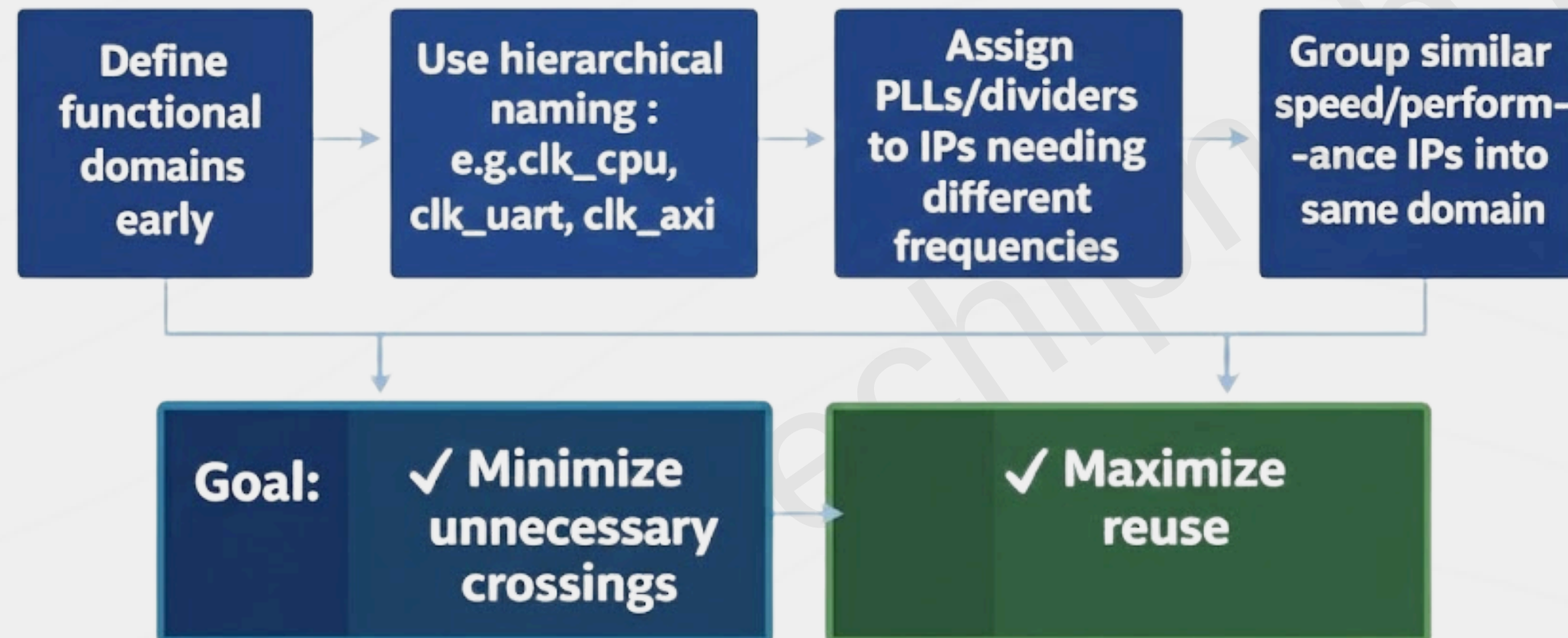


INTERFACE MATCHING



SAFE RESET/BOOT FLOW

➔ SoC Clock Domain Planning



Why Multiple Clock Domains Are Used

- Some IPs (e.g., USB, DDR) require fixed external clocks
- Reduces dynamic power by slowing unused blocks
- Supports asynchronous peripherals or interfaces
- Enables domain-specific clock gating or DVFS

Example:

- CPU@400MHz, UART@50MHz, SPI@25MHz

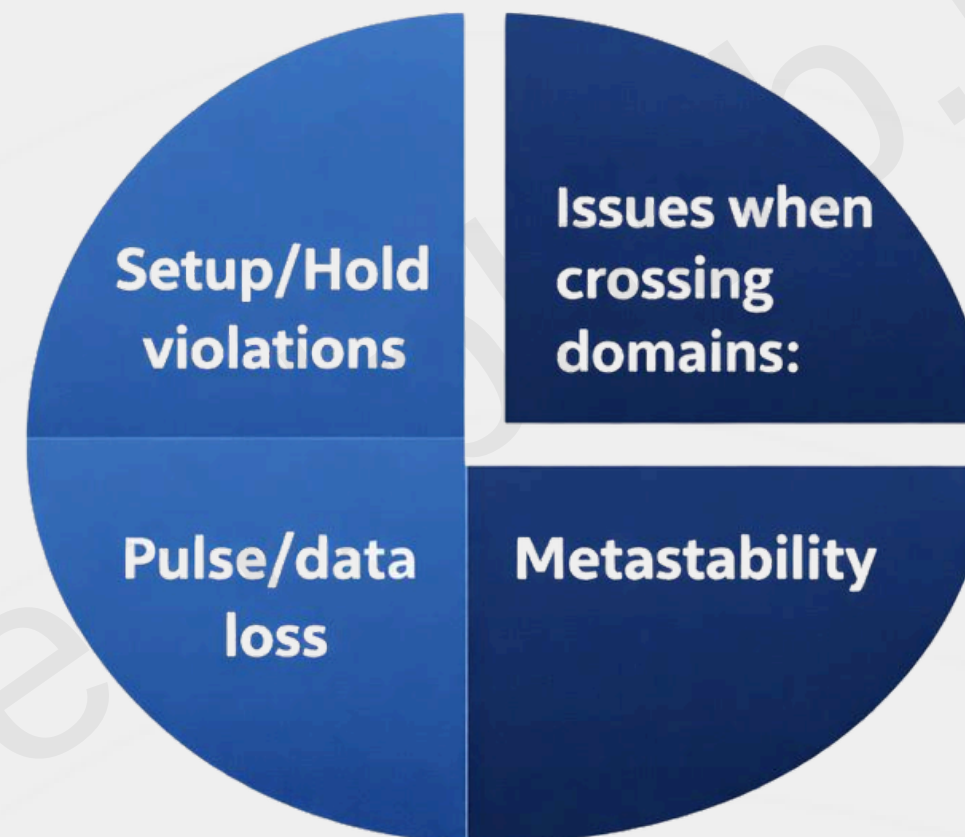
➔ Clock Domain Crossing Challenges Recap

Solution

- 2FF synchronizer (control)
- Async FIFO (data)
- Formal CDC checks

Tip:

- Use CDC-aware coding templates and simulation models



Reset Domain Partitioning and Design

1. Partition reset signals based on function and timing
2. Align reset control with clock domains
3. Avoid global reset-only logic

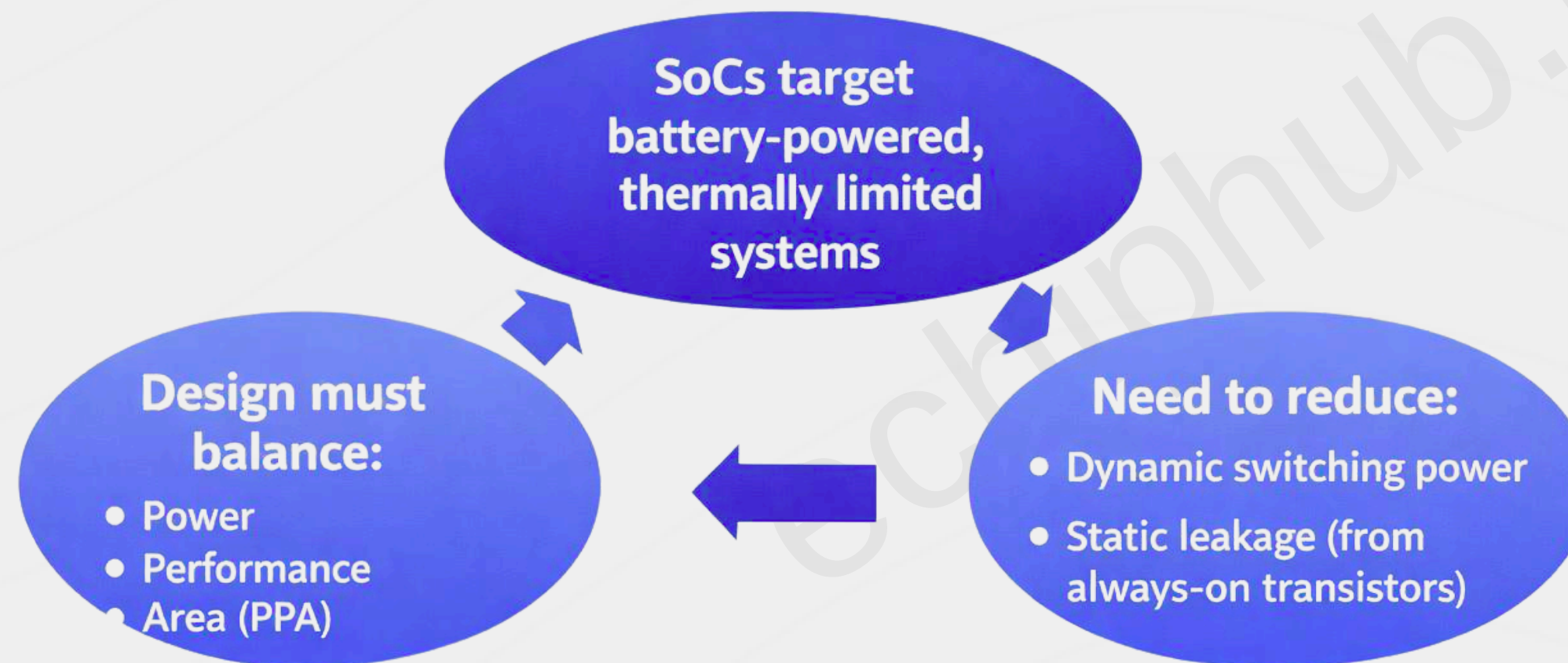
Some Reset domains:

- POR domain Triggers at boot; initializes base system)
- Debug reset :Used to restart subsystems without full reboot.
- IP-specific resets (e.g., uart_rstn) controls independent blocks locally.

➔ Global vs Local Reset Control

Global Reset:	Local Reset:	Best Practices:
• Comes from top-level controller • Affects all IPs simultaneously	• IP-specific • Controlled via status or FSM	• Reset synchronizers per domain • Avoid combinational resets • Use 'reset synchronizer' cells

➔ Low-Power Design – Challenges and Goals



➔ Dynamic Power vs Static Power

Dynamic Power:	Static Power:	Implication:
• From switching activity • Proportional to frequency $\times$ voltage² $\times$ activity	• From leakage currents in idle transistors • Increases with process scaling and temperature	• Use clock gating (dynamic) and power gating (static)

➔ Clock Gating and Power Gating Explained

Clock Gating:	Power Gating:	Guidelines:
• Stops clock to idle logic • Saves dynamic power	• Shuts off entire block (using power switches) • Requires isolation & retention logic	• Define power domains early • Simulate with power-aware testbenches

➔ Low-Power Design Methodologies (UPF/CPF)

UPF (Unified Power Format) / CPF (Common Power Format):	Used by:	Enables:
Describe power domains, strategies, retention	- Synthesis tools (e.g., DC, Genus) - Simulation tools (e.g., VCS, Questa)	- Power-intent modeling - Power-aware simulation and signoff

Summary - SoC-Level Best Practices

- Plan clock and reset domains during architecture
- Reduce domain crossings where possible
- Use CDC/RDC templates and checkers
- Apply clock gating for dynamic power savings
- Use power domains, UPF/CPF, and power gating
- Validate low-power flow early in RTL cycle
- Out come: Efficient, power-optimized, reliable SoC system

Thank You

echiphu

